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RESEARCH ARTICLE

Niacin prevents mitochondrial oxidative stress caused by sub-chronic exposure to methylmercury

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ABSTRACT

Humans and animals can be exposed to different chemical forms of mercury (Hg) in the environment. For example, methylmercury (MeHg)-contaminated fish is part of the basic diet of the riparian population in the Brazilian Amazon Basin, which leads to high total blood and plasma Hg levels in people living therein. Hg induces toxic effects mainly through oxidative stress. Different compounds have been used to prevent the damage caused by MeHg-induced reactive oxygen species (ROS). This study aims to investigate the *in vivo* effects of sub-chronic exposure to low MeHg levels on the mitochondrial oxidative status and to evaluate the niacin protective effect against MeHg-induced oxidative stress. For this purpose, Male Wistar rats were divided into four groups: control group, treated with drinking water on a daily basis; group exposed to MeHg at a dose of 100 μg/kg/day; group that received niacin at a dose of 50 mg/kg/day in drinking water, with drinking water being administered by gavage; group that received niacin at a dose of 50 mg/kg/day in drinking water as well as MeHg at a dose of 100 μg/kg/day. After 12 weeks, the rats, which weighed 500–550 g, were sacrificed, and their liver mitochondria were isolated by standard differential centrifugation. Sub-chronic exposure to MeHg (100 μg/kg/day for 12 weeks) led to mitochondrial swelling ($p < 0.05$) and induced ROS overproduction as determined by increased DFCH oxidation ($p < 0.05$), increased gluthatione oxidation ($p < 0.05$), and reduced protein thiol content ($p < 0.05$). In contrast, niacin supplementation inhibited oxidative stress, which counteracted and minimized the toxic MeHg effects on mitochondria.

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Methylmercury; mitochondria; niacin; protective effect; sub-chronic

Introduction

Mercury (Hg) is a well-known environmental pollutant that induces severe toxicological effects in biological systems (Stacey and Kappus 1982). The extent of damage varies according to Hg dosage, level of exposure to Hg, and Hg chemical form, but all Hg forms can be toxic (Roos *et al.* 2012). Humans and animals can be exposed to different Hg chemical forms in the environment, including elemental Hg vapor (Hg^0), inorganic Hg (i-Hg; mercurous ion [Hg^+], and mercuric ion [Hg^{2+}]), and organic Hg compounds like ethylmercury, methylmercury (MeHg), and dimethylmercury (Clarkson 2002, Bernhoft 2012). Humans can also be exposed to Hg through the use of medicinal products containing HgS, such as popular Tibetan medicines for sedation (Li *et al.* 2016), cosmetics, and skin-lightening creams (Borowska and Brzóska 2015, Agorku *et al.* 2016).

Results obtained between 2009 and 2011 for fish and seafood most commonly consumed by the Brazilian population living in different regions of Brazil (South, Southeast, and Northeast) showed that Hg levels were higher in fish (river and sea) than in marine invertebrates, which suggested that Hg bioaccumulates in the food chain (Grotto *et al.* 2012). Another study has reported high plasma MeHg and i-Hg concentrations in a population exposed to MeHg in the Brazilian Amazon Basin, where MeHg-contaminated fish is the basis of the diet of the riparian population, which leads to high total blood and plasma Hg levels (Passos and Mergler 2008, Pinheiro *et al.* 2008, Grotto *et al.* 2010, Carneiro *et al.* 2014). Moreover, studies have shown associations between environmental exposure to MeHg and increased oxidative stress, confirming that Hg adversely affects the riparian population (Pinheiro *et al.* 2008, Grotto *et al.* 2010).

One of the best-known mechanisms of Hg-induced toxicity is oxidative stress. Hg-induced oxidative imbalance routes primarily involve increased levels of reactive oxygen species (ROS), including hydrogen peroxide (H_2O_2), superoxide anion

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($O_2^{\bullet -}$), and hydroxyl radical ($OH^{\bullet}$) (Farina *et al.* 2011a), which are a consequence of impaired cell ROS scavenging potential resulting from high Hg affinity for thiol-containing molecules, such as reduced glutathione (GSH), and from Hg modulation of antioxidant enzymes like superoxide dismutase, catalase, and glutathione peroxidase (GSH-Px) (Clarkson 1997, Dringen 2000). In this context, different mechanisms and molecular targets have been proposed to explain MeHg toxicity, including thiol depletion, calcium deregulation, oxidative stress, mitochondrial dysfunction, and pyruvate transport (Aschner *et al.* 2007, Farina *et al.* 2011b, Grotto *et al.* 2011, Lee *et al.* 2016).

Hg can also deplete the GSH content and enhance ROS production, which then attack proteins (Halliwell and Chirico 1993), DNA (Marnett 1999), and mainly membrane lipids (Esterbauer and Cheeseman 1990). Even though there is no consensus regarding the behavior of enzymes involved in ROS scavenging after exposure to Hg, some studies have reported enzyme inhibition, transcription enhancement, and consumption, which can culminate in incompatible living conditions within the cell (Ariza *et al.* 1998, Bulat *et al.* 1998, Hussain *et al.* 1999, Chen *et al.* 2005). Other studies have shown that hydrogen sulfide (H_2S) and diphenyl diselenide administration attenuates MeHg effects both *in vitro* and *in vivo* (Glaser *et al.* 2014, Han *et al.* 2017).

The mitochondrial function is intrinsically linked to the mitochondrion structure. Any mitochondrial damage can trigger a cascade of events that impair essential cell functions (Pereira *et al.* 2012). Mitochondrion is an intracellular structure that transforms energy extracted from food into useful energy, which can then be transported by cells through adenosine triphosphate (ATP). Mitochondria are the primary source of ATP molecules, which are essential for life in eukaryotic cells (Bragadin 2006, Bernardi and Rasola 2007). Any change in mitochondrial function can disrupt ATP formation via oxidative phosphorylation as well as calcium homeostasis, which consequently impairs programed cell death (Martins de Brito and Scorrano 2010, Davies *et al.* 2011). Hence, mitochondria have become an important tool in the field of toxicology—they help to understand and to predict the adverse effects of numerous substances, and mitochondrial damage is often seen as a cause or a consequence of tissue injury (Leist *et al.* 1997, Wallace and Starkov 2000).

Different compounds have been used to prevent the damage caused by MeHg-induced ROS (Ornaghi *et al.* 1993, Heath *et al.* 2010, Sakamoto *et al.* 2013). For example, antioxidant compounds have been applied to avoid the damage resulting from ROS action (Ratnam *et al.* 2006).

Niacin (vitamin B3), also known as nicotinic acid, is an important antioxidant: it raises nicotinamide adenine dinucleotide (NAD) levels in tissues. NAD works as a coenzyme in cellular redox reactions, which makes it an essential component of metabolic pathways in all living cells (Hara *et al.* 2007). Niacin occurs in many foods, including fish, which are an important source of this vitamin (Berdanier 1998, Broad 1999).

This study aims to investigate the effects of sub-chronic exposure to low MeHg levels on rat liver mitochondria and to evaluate the possible niacin protective effect against Hg-induced oxidative stress.

Material and methods

Chemicals

MeHg, as methylmercury chloride (CH_3HgCl), and niacin were purchased from Sigma-Aldrich (St. Louis, MO, USA). Aqueous solutions were prepared in deionized water (Millipore, Bedford, MA, USA).

Reagents such as rotenone, carbonyl cyanide 3-chlorophenylhydrazone (CCCP), succinate, glutamate, malate, adenosine 5′-diphosphate sodium salt (ADP), o-phthalaldehyde (OPT), *N*-ethylmaleimide (NEM), ethyleneglycol bis (β-aminoethyl ether)-*N*,*N*,*N*′,*N*′-tetraacetic acid (EGTA), and 5,5′-dithiobis (2-nitrobenzoic acid) (DTNB) were purchased from Sigma-Aldrich Chemical Co (St Louis, MO, USA), while 2′,7′-dichlorodihydrofluorescein diacetate (H_2DCFDA) was obtained from Molecular Probes (OR, USA). All the other reagents used herein were of analytical grade. All stock solutions were prepared with glass-distilled deionized water.

Experimental treatment

Male Wistar rats from our Central Laboratory Animal Facility (University of São Paulo, Ribeirão Preto, Brazil) were used. The animals were kept under 12 h light/dark cycles in an air-conditioned room at 22–25 °C, with free access to food and water. Animal use complied with the guidelines of the Committee on Care and Use of Experimental Animal Resources of the University of São Paulo, Brazil (Process number 12.1.1599.53.8).

The rats were divided into four groups (two animals per group): (I) control group (treated with drinking water on a daily basis, by gavage); (II) group exposed to MeHg at a dose of 100 μg/kg/day by gavage (MeHg group); (III) group that received niacin in the drinking water at a dose of 50 mg/kg/day (Niacin group); (IV) group that received niacin in the drinking water at a dose of 50 mg/kg/day and MeHg at a dose of 100 μg/kg/day by gavage (MeHg + Niacin group).

The selected MeHg dose was based on studies about the daily MeHg ingestion by Brazilian riparian communities exposed to MeHg through contaminated fish intake. The choice of niacin dosage was based on literature data concerning the protective action of this compound even under oxidative stress conditions; hepatoprotection was also taken into account (Cho *et al.* 2009, Tupe *et al.* 2011). Exposure lasted 12 weeks, which is representative of sub-chronic exposure in rats.

Mitochondrial isolation

Rat liver mitochondria were isolated by standard differential centrifugation. Male Wistar rats weighing approximately 550 g were sacrificed by decapitation; their liver (10–15 g) was immediately removed and sliced into 50 mL of medium containing sucrose 250 mM, EGTA 1 mM, and HEPES–KOH 10 mM, pH 7.2, which was followed by homogenization in a

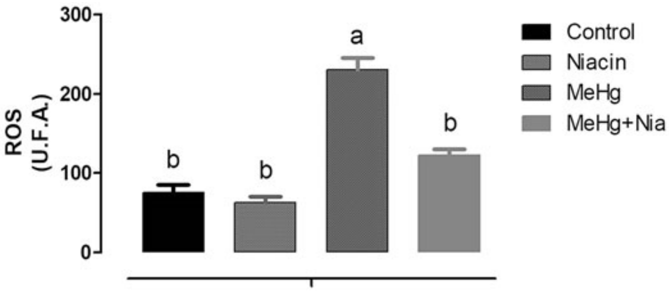

Figure 1. Effects of 12-week exposure to MeHg (100 μg/kg/day) and concomitant treatment with niacin (50 mg/kg/day) on ROS accumulation as assessed by using the fluorescence probe H_2DCFDA, The mitochondria were incubated in standard reaction medium containing sucrose 125 mM, KCl 65 mM, HEPES-KOH 10 mM, EGTA 0.5 mM, and K_2HPO_4 10 mM, pH 7.4. The data are expressed as the mean ± SEM. Means with the same letter do not differ statistically. Different letters correspond to statistically different values ($p < 0.05$).

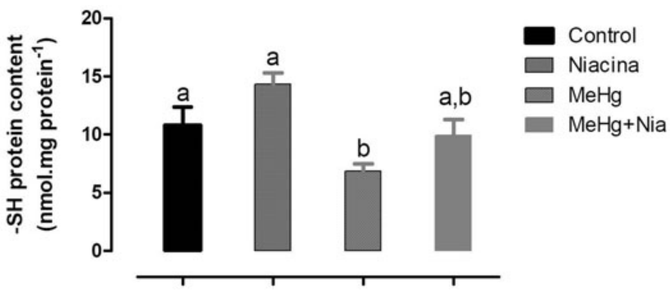

Figure 2. Effects of MeHg (100 μg/kg/day) and niacin (50 mg/kg/day) on mitochondrial protein sulfhydryl (PSH) content as assessed by using DTNB. The mitochondria were incubated in standard reaction medium containing sucrose 125 mM, KCl 65 mM, HEPES-KOH 10 mM, EGTA 0.5 mM, and K_2HPO_4 10 mM, pH 7.4. The data are expressed as the mean ± SEM. Means with the same letter do not differ statistically. Different letters correspond to statistically different values ($p < 0.05$).

Potter–Elvehjem homogenizer for 15 s at 1 min intervals, three times. The homogenate was centrifuged at 580 × *g* for 5 min, and the resulting supernatant was further centrifuged at 10,300 ×*g* for 10 min. The pellet was suspended in 10 mL of medium containing sucrose 250 mM, EGTA 0.3 mM, and HEPES–KOH 10 mM, pH 7.2, and centrifuged at 3,400 × *g* for 15 min. The final mitochondrial pellet was suspended in 1 mL of medium containing sucrose 250 mM and HEPES–KOH 10 mM, pH 7.2, and used within 3 h (Chance and Willians 1956). All the procedures were carried out at 4 °C, and the mitochondrial protein content was determined by the biuret reaction.

Mitochondrial swelling

Mitochondrial swelling was estimated from the decrease in apparent absorbance of 0.4 mg of protein at 540 nm. A spectrophotometer Model DU-70 spectrophotometer (Beckman, Coulter Inc., Fullerton, CA, U.S.A.) was employed. Cyclosporine-A 1 μM (CsA) and NEM 25 μM were used as modulators of the process in order to improve understanding of the mechanisms that trigger the process (Lemasters 1999).

Redox state analysis

Mitochondrial ROS production was spectrofluorimetrically assessed on a fluorescence spectrophotometer Model F-4500

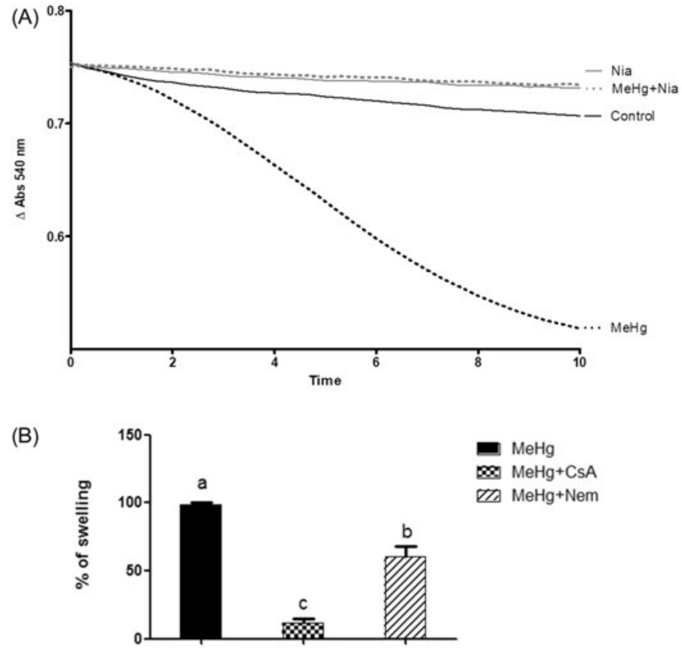

Figure 3. Effects of 12-week exposure to MeHg (100 μg/kg/day) and concomitant treatment with niacin (50 mg/kg/day) on mitochondrial swelling (A), and effects of the modulators Cyclosporin A (CsA) and NEM on the effects of the exposure to MeHg (B). The mitochondria were incubated in a standard reaction medium containing sucrose 125 mM, KCl 65 mM, HEPES-KOH 10 mM, EGTA 0.5 mM, and K_2HPO_4 10 mM, pH 7.4. The data are expressed as the mean ± SEM. Means with the same letter do not differ statistically. Different letters correspond to statistically different values ($p < 0.05$).

Hitachi; H_2DCFDA, the 503/529 nm excitation/emission wavelength pair, and 1 mg of protein/mL were used (Ames 1983). *Reduced (GSH) and Oxidized (GSSG) glutathione* were determined in the liver homogenates by the fluorimetric *ortho*-phthalaldehyde method (Hissin and Hilf 1976). *Protein Thiol* (PSH) *concentration* was evaluated after treatment of 1 mg of protein/mL with perchloric acid (5% final concentration) and centrifugation at 3400 ×*g* for 10 min. The pellet was suspended in 1 mL of medium containing EDTA 50 mM and Tris100 mM, pH 8.0. Next, DTNB 2 mM was added to the solution, and the absorbance was determined with the aid of a spectrophotometer Model DU-70 (Beckman, Coulter Inc., Fullerton, CA, U.S.A.) at 412 nm. The amount of thiol groups was calculated by using $\varepsilon = 13600\,M^{-1}\,cm^{-1}$ (Jocelyn 1987).

Data analysis

Experiments were performed in duplicate for each preparation. The averages and the respective standard deviations were calculated. The data are representative of each group and are expressed as the mean ± SEM (standard error mean). ANOVA (non-parametric) and *post-hoc* Tukey were used to perform the statistical analyses; GraphPad Prism Software, version 5.0 (GraphPad Software, USA), was employed. Values of $p < 0.05$ were considered significant.

Results

Because several studies have reported that oxidative stress induces cytotoxicity (Kowaltowski and Vercesi 1999, Costa

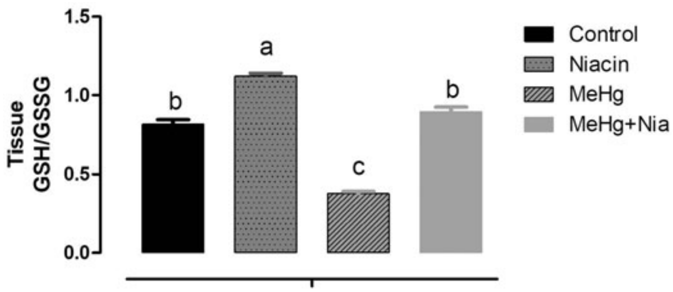

Figure 4. Effects of MeHg (100 μg/kg/day) and niacin (50 mg/kg/day) on the GSH/GSSG ratio analyzed by using the fluorescence probe OPT. The mitochondria were incubated in standard reaction medium containing sucrose 125 mM, KCl 65 mM, HEPES-KOH 10 mM, EGTA 0.5 mM, and K_2HPO_4 10 mM, pH 7.4. The data are expressed as the mean ± SEM. Means with the same letter do not differ statistically. Different letters correspond to statistically different values ($p < 0.05$).

et al. 2011, Lenaz 2012), we assessed the redox state of mitochondria to confirm whether sub-chronic exposure to MeHg actually increases ROS and to evaluate whether niacin prevents any oxidative damage caused by exposure to MeHg. Figure 1 shows that exposure to MeHg for 12 weeks considerably raised the mitochondrial ROS levels. Concomitant treatment with niacin decreased ROS production and prevented oxidative stress.

MeHg has great affinity for sulfhydryl groups (Farina *et al.* 2011b). On the basis of Figure 2, the MeHg group oxidized the sulfhydryl groups while Niacin co-administration partially inhibited mitochondrial thiol oxidation. (Figure 2). The oxidation of sulfhydryl groups may have destabilized the mitochondrial membrane and opened the mitochondrial permeability transition (MPT) pore, as illustrated in Figure 3, which shows that niacin supplementation fully protected against the mitochondrial swelling induced by exposure to MeHg. (Figure 3(A)). Moreover, in the presence of NEM, a monofunctional thiol reagent, swelling was partially (40%) and significantly inhibited. CsA inhibited mitochondrial swelling by 70% (Figure 3(B)).

The oxidative stress induced by MeHg exposure sufficed to prompt an imbalance in the tissue antioxidant defense. Figure 4 depicts the decrease in the GSH/GSSG ratio in the liver of rats exposed to MeHg. Niacin significantly increased the GSH/GSSG ratio as compared to the control group, and it protected the endogenous antioxidant defense from the oxidative stress induced by MeHg.

Discussion

Much has been learned about the neurotoxicity of MeHg recently (Toimela and Tahti 2004, Yin *et al.* 2007, Belyaeva 2010, Mori *et al.* 2011). Although it is known that MeHg rapidly accumulates in the liver and is eliminated through the bile (Takahashi *et al.* 1971, Chen *et al.* 2006), data about the effects of this metal are still lacking.

This work has established that sub-chronic exposure to MeHg causes liver mitochondrial damage by increasing mitochondrial membrane permeability due to oxidative stress, which is reflected in the tissue glutathione levels. This is expected because the oxidative damage promoted by MeHg occurs via its reaction with thiol (–SH) and/or selenol (–SeH) groups of endogenous molecules (Farina *et al.* 2011b). Selenol groups are stronger and softer nucleophiles than thiol groups, so MeHg affinity for selenol is higher as compared to an analog thiol (Farina *et al.* 2011b, Farina *et al.* 2011c). To our knowledge, this is the first study that has investigated MeHg-induced toxicity in rat liver mitochondria by conducting *in vivo* sub-chronic exposure of rats to MeHg. This is also the first study that has demonstrated that niacin co-administration can counteract the mitochondrial damage elicited by Me-Hg and prevent the oxidative stress caused by exposure to MeHg. Other studies have also reported mitochondrial dysfunction due to exposure to MeHg. Lee *et al.* (2016) showed that exposure of human neuroblastoma cells to MeHg decreases the mitochondrial membrane potential, and that pyruvate addition significantly diminishes the potential even further, whereas treatment with an antioxidant inhibits the MeHg effects. Similarly, H_2S or diphenyl diselenide $[(PhSe)_2]$ addition attenuates MeHg-induced neurotoxicity in rats (Dalla *et al.* 2013, Han *et al.* 2017). MeHg may cause neuronal damage, with special emphasis on mitochondrial disorders and glutamate and calcium dyshomeostasis (Roos, *et al.* 2012).

Mitochondrial energy metabolism (oxidative phosphorylation) is currently recognized as the predominant ROS source in most eukaryotic cell types (Kowaltowski *et al.* 2009, Sinha *et al.* 2013). When ROS levels are increased and/or antioxidant defenses are compromised, mitochondria are more susceptible to damage (Kowaltowski and Vercesi 1999, Pessayre *et al.* 2012). This condition is defined as Oxidative Stress, an imbalance between antioxidants and pro-oxidants that favors pro-oxidants (Sies 1985, Kehrer and Lund 1994) and has been implicated in the toxicity of many xenobiotics (Wallace and Starkov 2000). ROS production by mitochondria may accelerate under various pathological conditions; for example, hypoxia, ischemia, and reperfusion; interaction with chemical compounds also accelerates ROS production in mitochondria (Chen *et al.* 2003). Indeed, numerous studies have shown that electron transport chain inhibition is associated with increased ROS production (Chen *et al.* 2003, Adam-Vizi 2005, Monaca and Fodale 2012).

In this context, several reports have demonstrated electron transport chain inhibition by MeHg. For example, MeHg inhibited the mitochondrial electron transport chain in PC12 cell culture exposed to MeHg 10 μM for 3 h and in mitochondria isolated from cerebellum of rats exposed to 10 mg of MeHg/kg for five consecutive days (Mori *et al.* 2011, Belyaeva *et al.* 2012). These findings indicate that electron transport chain inhibition is a common toxic effect of exposure to MeHg. Proper electron transport chain functioning is vital to mitochondrion and tissue integrity and function (Kilpatrich-Smith *et al.* 1981).

Our experimental model clearly shows signs of redox imbalance resulting from mitochondrial injury. As reported by Yin *et al.* (2007) MeHg induces oxidative stress due to increased ROS production. Our findings corroborate this by showing that exposure to MeHg considerably raises ROS levels. However, concomitant niacin administration can protect the liver against the damage induced by MeHg. Although niacin supplementation cannot completely eradicate the ROS, results are statistically different from the results obtained for the group exposed to MeHg only, and analyses of other end points, such as PSH oxidation and GSH/GSSG ratio in the tissue reveal even higher niacin efficacy.

Under oxidative stress conditions, isolated mitochondria may experience the phenomenon known as MPT (Lapidus and Sokolove 1993). On the basis of our results, sub-chronic exposure to MeHg also opens the permeability transition pore. This process is characterized by mitochondrial swelling, which is sensitive to inhibition by several modulators, such as CsA (Zoratti and Szabò 1995). This may be one of the aggravating mechanisms of MeHg toxicity because MPT is directly associated with cell death induction, which can be a cause or a consequence of oxidative stress.

This study has shown that long-term exposure to MeHg induces mitochondrial swelling under the tested conditions. Furthermore, CsA, a specific MPT inhibitor, completely abates mitochondrial swelling, which suggests that swelling is a result of the MPT process and is only partially inhibited by NEM, a monofunctional thiol reagent.

NEM prevents mitochondrial swelling by protecting sulfhydryl groups because it can remove GSSG (Guntherberg and Rost 1966). However, the NEM partial ability to protect against swelling after exposure to MeHg indicates that the observed MPT originates from MeHg-induced oxidative stress.

This study complements and is in line with other previous studies that have demonstrated that niacin exerts hepatoprotective effect under oxidative damage conditions (Yan *et al.* 1999, Cho *et al.* 2009, Ganji *et al.* 2009). In addition, niacin has been shown to have potent antioxidant and anti-inflammatory properties (Ganji *et al.* 2009, Tupe *et al.* 2011). Therefore, we also propose that niacin antioxidant properties come into play to decrease MeHg-induced oxidative stress: niacin preserves the tissue antioxidant defense, as assessed by the GSH/GSSG ratio. In this context, the main source of reducing equivalents for antioxidant systems such as glutathione peroxidase/reductase and thioredoxine peroxidase/reductase is NADPH (Hoek and Rydstrom 1988). As a precursor of NAD^+ synthesis, niacin raises the cellular NAD^+ concentration and consequently upregulates the expression of glucose-6-phosphate dehydrogenase, one of the main sources of NAD(P)H (Tupe *et al.* 2011).

Niacin administration attenuates ROS production and increases the GSH/GSSG ratio, thereby preventing lipid and protein oxidation, as evidenced by inhibited GSH depletion and reduced sulfhydryl protein oxidation. That is an important issue to the Amazon population, who are chronically exposed to MeHg through daily fish consumption (Passos and Mergler 2008); while niacin, a fish nutrient, may provide a protective effect against exposure to MeHg. In view of the beneficial niacin effects after exposure to MeHg, a potential treatment for Hg intoxication may emerge given that niacin addition mitigates the genotoxic effects produced by MeHg (de Paula *et al.* 2016).

In conclusion, daily long-term exposure to low MeHg doses is sufficient to establish mitochondrial injury and adversely impact the liver of exposed rats. In contrast, long-term niacin supplementation can mitigate the oxidative stress scenario. Mitochondrial redox state preservation is very important for proper functioning of the organelle because it avoids cell death and tissue failure resulting from exposure to MeHg. Taken together, our findings demonstrate significant damage in rat liver mitochondria after exposure to low MeHg levels and show the important protective profile of niacin, which can counteract and minimize the MeHg effects.

Disclosure statement

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